Tuesday Warm Up: ADCs, and Digital to Analog Converters (DACs)

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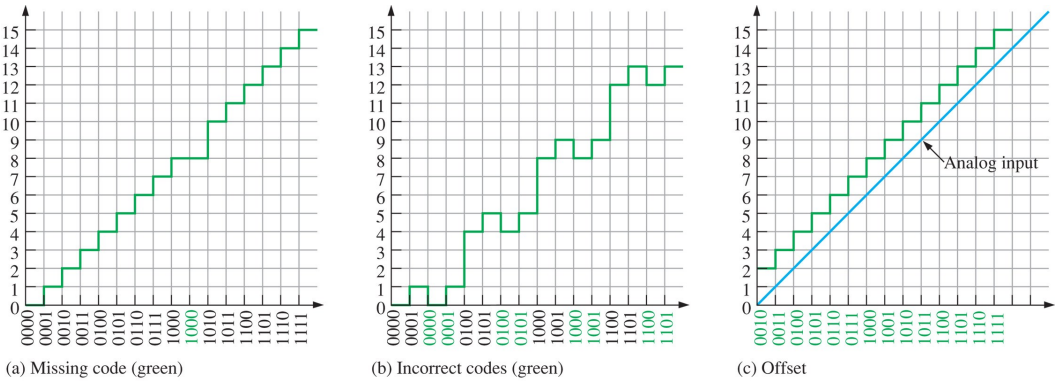


Figure 1: Types of ADC errors.

1 Conversion Errors

1. (a) Consider Fig. 1 (a). What analog voltage should go with the ADC output 1000? (b) Consider Fig. 1 (b). What ADC code should go with the ADC input 15 V? (c) Consider Fig. 1 (c). What is the binary error that is ocurring relative to the analog voltage input?

2. (a) Consider Fig. 2 (b). In a DAC, a user inputs 1111, but obtains 11 V instead of 15 V. What is the voltage error for the code 1001? (b) Consider Fig. 2 (c). What are the slopes of the three lines, in volts per count?

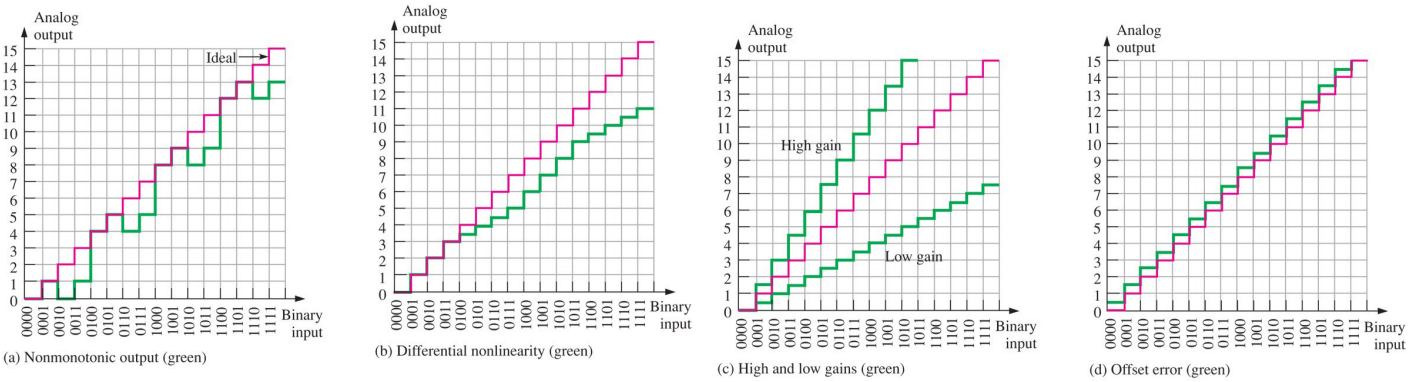


Figure 2: (a) DAC coding errors. (b) DAC non-linearity. (c) DAC gain errors. (d) DAC offset errors.